

Rutwik Patel

San Francisco, CA | (213) 913-8803 | me.rutwik@gmail.com | LinkedIn/rutwikpatel13 | GitHub/RutwikPatel13

EDUCATION

University of Southern California, Los Angeles, CA

August 2023 - May 2025

Masters of Science in Computer Science

GPA: 3.81/4.0

University of Mumbai, Mumbai, India

August 2019 - May 2023

Bachelor of Technology in Information Technology

GPA: 3.8/4.0

EXPERIENCE

USC Marshall School of Business

Los Angeles, CA

Research Assistant

Jan 2026 - May 2026

- Built a **RAG-based** semantic search and chat platform over **10K+ vector embeddings** that lets researchers query **500GB+** of structured and unstructured sustainability disclosures directly in natural language
- Developed chunking, schema-guided **LLM pipelines** that extract structured ESG fields from unstructured disclosures
- Shipped a **React** dashboard and chat interface for **30+ researchers**, cutting **15+ hours per week** of manual review
- Integrated an **LLM layer** that grounds answers in retrieved disclosures with **citations** to reduce hallucination
- Added **feedback loop** capturing researcher ratings and corrections to refine retrieval and improve accuracy over time

Research Assistant

Feb 2024 - May 2025

- Engineered fault-tolerant **ETL pipelines** chaining **Selenium** scraping, OCR, and indexing to ingest **15,000+** sustainability reports across 10 years of S&P 1500 filings into a unified research knowledge base
- Modeled normalized **PostgreSQL** schemas with targeted indexing over **2M+** extracted data points, cutting query latency by 12% and powering fast downstream analytics, semantic retrieval, and reporting across the corpus
- Parallelized PDF extraction with **multithreading**, cutting processing time **30%** over the prior sequential pipeline

Software Engineer Intern

Sept 2025 - Dec 2025

Sigma Computing

New York, NY

- Delivered **four production features** end-to-end owning implementation, testing, and deployment including a condition-based formula visualization tool, Form_v2, headers, navigation used by **60+ enterprise organizations** within a week
- Optimized rendering of large data grids with **memoization and virtualization**, improving dashboard responsiveness
- Wrote **Cypress** end-to-end tests for critical workbook flows, catching regressions before they reached production

Software Engineer

Jul 2024 - Aug 2025

World Salon

Los Angeles, CA

- Launched an events platform powering **130+ events** for institutions, speakers, attendees, BDR and internal teams
- Engineered scraping and **OpenAI**-powered profiling pipelines that processed **58,000+** candidate profiles end-to-end, automating the company's core speaker-sourcing operation and replacing hours of manual research and outreach
- Designed **REST APIs** with **JWT** auth and **role-based access control** across production services and admin routes
- Refactored a monolithic backend into modular payment and event services, reducing request latency by **20%**
- Containerized and deployed applications on **AWS** using **Docker**, EC2, S3, and **GitHub Actions CI/CD** pipelines

ACADEMIC PROJECTS

miniredis – Redis-Compatible In-Memory Store (GitHub) | Built a Redis-compatible store in **C++** with a non-blocking **epoll/kqueue** reactor over the RESP protocol, backing sorted sets with a hand-written skip list and reaching 984K+ ops/sec, **2.5x** higher throughput via sharded locking, load-tested with a **Go** benchmark harness

TalkToData – Natural-Language SQL (Demo, GitHub) | Created an AI platform converting natural-language questions into validated **SQL** across databases, using schema-aware prompting and a **secure execution layer** that blocks unsafe queries; provisioned **AWS** infrastructure (EC2, RDS, S3) as code with **Terraform** for reproducible deploys

RoomReserve (Demo, GitHub) | Designed **20+ REST endpoints** for auth, booking lifecycle, and payments on a normalized relational schema, with role-based access control and **concurrency-safe** double-booking prevention.

Cataract Detection with Explainable AI (Demo, GitHub) | Trained an interpretable CNN reaching **97%** accuracy on retinal images with **Grad-CAM** heatmaps that visualize model attention and highlight affected regions through a clinician-facing web interface that returns predictions with confidence scores; published as an **IEEE conference paper**.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C++, Go | **Databases:** PostgreSQL, MongoDB, Redis, Chroma

Web Development: Node, React, Django, Flask, Docker, Kubernetes, CI/CD, Jest, Cypress

Cloud & DevOps: AWS, GCP, Docker, CI/CD, GitHub Actions, Terraform, Monitoring & Logging